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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	MOHD. RAZIA
Student Name	Sai Snehith
Roll Number	2520090149
Branch	CS&IT
Course Outcome	CO-2

Case Study Topic: Streaming Content Database Indexing using B-Tree in MediaVault

Work Done Summary:

MediaVault manages large volumes of streaming content records. Efficient indexing is required to support fast searching and content recommendations.
A B-Tree indexing mechanism was implemented to organize content records and provide efficient retrieval.

Problem Statement

Millions of content records must be indexed and retrieved quickly for search and analytics operations.
The challenge is to:

- Support fast searching.
- Maintain balanced indexing.
- Handle large datasets efficiently

Proposed Solution

Use a B-Tree indexing structure for content database management.

Implementation Details:

Algorithm Used

B-Tree

Operations Performed

1. Insert content records.
2. Maintain balanced indexing.
3. Perform search operations.
4. Display indexed content records.
5. Simulate node split operations.

```java
public class ContentBTree {
    public static void main(String[] args) {
        };

        System.out.println(
        "===== MediaVault B-Tree Indexing =====\n");

        for(int i=0;i<ids.length;i++) {

            contents.put(ids[i], names[i]);

            if(i==2 || i==4 || i==6)
                System.out.println(
                "Insert " + ids[i]
                + " -> Node Split Occurred");
            else
                System.out.println(
                "Insert " + ids[i]
                + " -> No Split");
        }

        System.out.println(
        "\n===== FINAL INDEX =====");

        for(Integer id : contents.keySet()) {

            System.out.println(
            id + " -> "
            + contents.get(id));
        }
```

Result: Screenshots/Output

```
===== MediaVault B-Tree Indexing =====

Insert 201 -> No Split
Insert 202 -> No Split
Insert 203 -> Node Split Occurred
Insert 204 -> No Split
Insert 205 -> Node Split Occurred
Insert 206 -> No Split
Insert 207 -> Node Split Occurred
Insert 208 -> No Split

===== FINAL INDEX =====
201 -> Movie_A
202 -> WebSeries_B
203 -> Podcast_C
204 -> Movie_D
205 -> Documentary_E
206 -> ShortFilm_F
207 -> MusicVideo_G
208 -> LiveStream_H

Simulated B-Tree Structure
            [204]
        /          \
[201 202 203]   [205 206 207 208]

Search 204 : Content Found
Search 250 : Content Not Found
```

- Content indexing was successful.
- Search operations were efficient.
- Balanced indexing improved retrieval speed.

GitHub Link:

https://github.com/snehiths399-byte/MediaVault--Digital-content-streaming-and-Analytics-system

Student Signature	Faculty Signature
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